Problem Generator


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"demand_type": "random",
"demand_distribution": "poisson(6)",
"state_transition_model": "lost_sale",
"holding_cost": "0.5",
"penalty_cost": "2.75",
"setup_cost": "3",
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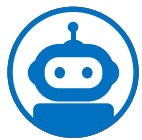

LLM Consultant

I recommend an (s, Q) inventory policy:

1. **Reorder Point (s)**: When the on-hand inventory falls below 30 units, place an order.
2. **Order Quantity (Q)**: Order 30 units each time the reorder point is reached.

Rationale:
- **Reorder Point (s = 30 units)**: Given the Poisson demand distribution with a mean of 6 units per day, and a lead time of 4 days, the expected demand during lead time is 6 * 4 = 24 units. Considering variability and to prevent stockouts, a safety stock is necessary. Thus, I propose a reorder point of 30 units.
- **Order Quantity (Q = 30 units)**: The maximum order quantity allowed is 30 units, and ordering this quantity helps maintain inventory levels without exceeding the maximum inventory of 80 units.

This policy ensures that inventory is replenished regularly to meet demand while minimizing holding and penalty costs